Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Responsible Generative AI Basics

TGS Ref No: TGS-2025060472

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 3.1

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	28 August 2025	First version.	Tertiary Infotech Academy Pte Ltd
2.0	17 October 2025	Update company name.	Tertiary Infotech Academy Pte Ltd
3.0	11 August 2026	Courseware rebuilt on the single-source WSQ pipeline: slide deck redesigned in the all-white house style with visual components (tile grids, comparison tables, flow diagrams, stat bands) replacing imported screenshots; the 14 in-class web activities documented as step-by-step labs; Lesson Plan and Learner Guide regenerated and aligned to the deck.	Tertiary Infotech Academy Pte Ltd
3.1	11 August 2026	QA fixes: Lesson Plan slide citations corrected to the true deck index and the daily schedule rebalanced so the stated hours reconcile; Learner Guide gains a UI screenshot and a 'Test it' verification box for each of the 14 activities; the 7-step governance checklist redrawn as a tile grid to stop chip text clipping.	Tertiary Infotech Academy Pte Ltd

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Course Information

Course Title	Responsible Generative AI Basics
WSQ Course Reference	TGS-2025060472
TSC Reference	Responsible AI and Generative AI Practices (ICT-BAS-0055-1.1) · Level 1
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	1 day · 9:30am–7:05pm (7.67 instructional hours + 1 hour of assessment)
Daily Timing	45-minute lunch; two 10-minute tea breaks counted within training time
Mode	Instructor-led, with case-study and workshop activities per Learning Unit
Instructional Methods	Interactive presentation, discussions, case studies, peer teaching / peer practice
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Apply ethical reasoning to assess generative AI outputs and guide responsible decision-making in AI implementation.
- LO2: Apply privacy safeguards to protect sensitive data used in generative AI through anonymisation and secure storage methods.
- LO3: Compare generative AI systems for ethical compliance based on intellectual property, privacy, and environmental impact.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 30 minutes, open book.
- Case Study (CS) — one continuous scenario with open-ended tasks, 30 minutes, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- The assessment block runs at the end of the training day, from 6:05pm to 7:05pm.
- The Written Assessment covers the knowledge statements K1–K4; the Case Study covers the ability statements A1–A6. A candidate must be Competent in every K and A.
- A minimum of 75% attendance (per SSG Digital Attendance record) is required to be eligible for assessment and funding. Learners must be assessed as Competent in every K and A, and complete the TRAQOM survey.

Daily Schedule

Day 1 — Ethical Principles, Privacy Techniques, Responsible AI Best Practices & Assessment

Time	Duration	Topic / Activity	Slides
9:30–9:45	15 min	Digital Attendance (AM), welcome, trainer and learner introductions, ground rules, Skills Framework and course outline	—
9:45–10:20	35 min	**LU1 Topic 1: Ethical considerations and potential risks of generative AI interaction (K2) — hidden costs, deepfakes, environmental and societal impact, the deconstruction of AI bias**	Slide 18
10:20–10:40	20 min	Activity 1: AI Ethical Dilemma Simulator	Slide 30
10:40–11:05	25 min	**LU1 Topic 2: Ethical principles in AI (K4) — AI Verify's 11 principles, the GenAI risk diagnostic matrix, prompt-injection threat model, Project Moonshot**	—
11:05–11:15	10 min	Tea break	—
11:15–11:45	30 min	Activity 2: AI Ethical Principles Quiz and Activity 3: Prompt Injection Playground	Slides 31–32
11:45–12:10	25 min	**LU1 Topic 3: Apply ethical principles in decision-making related to AI (A1) — three pillars of algorithmic risk, national AI policy, the EU AI Act compliance waterfall, human in the loop**	—
12:10–12:55	45 min	Lunch break	—
12:55–1:20	25 min	Activity 4: Ethical Decision-Making Framework	Slide 33
1:20–1:40	20 min	**LU1 Topic 4: Exercise professional**	—

		scepticism to exercise sound judgement on GenAI output (A2) — automation bias and the vulnerability in the human filter	
1:40–2:10	30 min	Activity 5: AI Output Scepticism Checker and Activity 6: The Cognitive Threat Matrix; LU1 recap	Slides 34–35
2:10–2:20	10 min	Digital Attendance (PM)	—
2:20–2:45	25 min	**LU2 Topic 1: Data anonymisation and de-identification techniques (K1) — anonymisation vs pseudonymisation, k-anonymity, choosing the right protection**	Slide 38
2:45–3:05	20 min	Activity 7: Data Anonymisation Techniques	Slide 47
3:05–3:30	25 min	**LU2 Topics 2–3: Apply privacy measures when handling data for AI applications (A3) and design guidelines for ethical AI use of sensitive data (A4) — the three layers of AI data security, the 7-step governance checklist**	—
3:30–3:40	10 min	Tea break	—
3:40–4:10	30 min	Activity 8: Python Data Anonymisation Pipeline and Activity 9: Ethical AI Data Policy Generator; LU2 recap	Slides 48–49
4:10–4:35	25 min	**LU3 Topics 1–2: Responsible AI principles and best practices, and comparing AI systems on IP, data privacy and environmental impact (K3, A5) — EU AI Act risk taxonomy, ISO/IEC 42005, differential privacy, Red AI vs Green AI**	Slide 52
4:35–5:05	30 min	Activities 10–12: AI	Slides 65–67

		System Comparison Matrix, Differential Privacy Explorer and Explainable AI (XAI) Explorer	
5:05–5:20	15 min	**LU3 Topic 3: Compare ethical issues in AI applications (A6) — the anatomy of algorithmic discrimination, accuracy vs interpretability, LIME vs SHAP**	—
5:20–5:45	25 min	Activities 13–14: The Ethical Paradigms of AI Resource Allocation and AI Ethics Case Study Analyzer; LU3 recap	Slides 68–69
5:45–5:55	10 min	Course summary, what you achieved, Q&A and Courseware & Assessment on the LMS	—
5:55–6:05	10 min	**Briefing for Assessment and Digital Attendance (Assessment)**	Slide 12
6:05–6:35	30 min	**Written Assessment (WA) — Short-Answer Questions (SAQ), 30 minutes, open book (K1–K4)**	Slide 13
6:35–7:05	30 min	**Case Study (CS) — 30 minutes, open book (A1–A6). TRAQOM survey and course feedback**	Slide 74

Instructional time: 460 minutes (7.67 hours), excluding the 45-minute lunch.   Assessment block: 70 minutes (briefing + WA + CS).   Total contact time: 530 minutes (8.83 hours).

The assessment itself is 1 hour — Written Assessment (SAQ) 30 minutes plus Case Study 30 minutes — in line with Assessment Plan v2.0 (0.5 hrs per instrument, 1 hr total), preceded by a 10-minute assessment briefing. Digital attendance is taken three times — AM, PM and Assessment — as required for WSQ-funded courses.

Topic-by-Topic Breakdown

LU1 — Ethical Principles of Generative AI · Slide 18

Risks of GenAI Interaction · Ethical Principles · Ethical Decision-Making · Professional Scepticism

Key concepts:

- Generative AI carries hidden costs beyond the subscription fee: environmental load (energy, water, carbon), security exposure, and psychological or societal impact on the people who use it.
- Deepfakes, voice cloning and face-swapping have made synthetic media cheap and convincing — eroding the assumption that seeing is believing.
- AI bias is not a single defect: it is introduced through the training data, the model objective, and the deployment context, and it compounds at each stage.
- Singapore's AI Verify framework defines 11 principles of trustworthy AI, split into process checks and technical tests, and is operationalised through Project Moonshot.
- Applying ethics in practice means stratifying risk (EU AI Act), keeping a human in the loop for high-stakes decisions, and engineering ethics into the design, not bolting it on.
- Professional scepticism is the discipline of verifying GenAI output against source evidence — the antidote to automation bias, where people defer to a confident machine.

Activity 1 — AI Ethical Dilemma Simulator · Slide 30

Objective: Recognise the ethical considerations and potential risks that arise when interacting with generative AI (K2).

Scenario: Work through six real-world AI dilemmas — from AI-generated hospital discharge summaries to resume-screening bias — choosing the most defensible course of action in each and reading the consequences of the choice you made.

Duration: 10 minutes.

Activity 2 — AI Ethical Principles Quiz · Slide 31

Objective: Identify and apply the core ethical principles in AI (K4).

Scenario: Test your grasp of the principles that underpin trustworthy AI — fairness, transparency, accountability, privacy, beneficence and human agency — across ten applied questions.

Duration: 10 minutes.

Activity 3 — Prompt Injection Playground · Slide 32

Objective: Explain how prompt injection threatens AI safety and how layered defences reduce the risk (K2).

Scenario: Attack three simulated AI assistants — an undefended customer-service bot, a banking assistant with basic defences and a medical triage AI with strong defences — and see which injections get through each layer of protection.

Duration: 10 minutes.

Activity 4 — Ethical Decision-Making Framework · Slide 33

Objective: Apply ethical principles systematically in decision-making related to AI (A1).

Scenario: Take one contested AI deployment through a six-stage structured ethical analysis — scenario, stakeholders, risks, principles, alternatives, decision — and produce a defensible, documented recommendation rather than a gut reaction.

Duration: 15 minutes.

Activity 5 — AI Output Scepticism Checker · Slide 34

Objective: Exercise professional scepticism to exercise sound judgement on generative AI output (A2).

Scenario: Evaluate confident, fluent, and quietly wrong AI-generated text against a 16-point scepticism checklist, and score how far it can actually be trusted.

Duration: 10 minutes.

Activity 6 — The Cognitive Threat Matrix · Slide 35

Objective: Analyse how human cognitive bias combines with generative AI failure modes to escalate risk (K2, A2).

Scenario: Pair a human cognitive bias with the AI failure mode it amplifies — automation bias with hallucination, confirmation bias with skewed data — and generate a structured risk analysis of a scenario from your own workplace.

Duration: 15 minutes.

LU2 — Generative AI Privacy Techniques · Slide 38

Anonymisation & De-identification · Privacy Measures in Practice · Ethical Data Guidelines

Key concepts:

- Anonymisation irreversibly severs the link to an individual; pseudonymisation only replaces identifiers and remains personal data under the PDPA and GDPR.
- K-anonymity guarantees an individual cannot be distinguished from at least k-1 others by generalising or suppressing quasi-identifiers.
- Release-based privacy fails when an attacker holds external auxiliary data — the re-identification attack that motivated stronger, computation-based guarantees.
- The three layers of AI data security are the pipeline (encryption in transit and at rest), the model (training and memorisation controls) and the output (filtering).
- Format-preserving encryption and data-shifting keep a field usable by downstream systems while removing its identifying power.
- An ethical AI data policy classifies data as prohibited, restricted or permitted for AI input, and pairs each class with a mandated safeguard.

Activity 7 — Data Anonymisation Techniques · Slide 47

Objective: Distinguish the anonymisation spectrum — de-identification, pseudonymisation, objective and subjective anonymisation (K1).

Scenario: Run one set of sensitive records through four different anonymisation techniques side by side, and see why only some of them actually remove the re-identification risk.

Duration: 15 minutes.

Activity 8 — Python Data Anonymisation Pipeline · Slide 48

Objective: Apply privacy measures when handling data for AI applications by building an anonymisation pipeline (A3).

Scenario: Build a seven-stage anonymisation pipeline over a raw dataset — pseudo-IDs, text shifting, masking, generalisation and column dropping — then export both the anonymised data and the Python code that produced it.

Duration: 15 minutes.

Activity 9 — Ethical AI Data Policy Generator · Slide 49

Objective: Design guidelines or strategies for ethical AI use, including use of sensitive data in generative AI tools (A4).

Scenario: Produce a complete, organisation-specific AI data-handling policy through a guided five-step wizard — the document that tells your colleagues what they may and may not paste into an AI tool.

Duration: 15 minutes.

LU3 — Best Practices of Responsible Generative AI · Slide 52

Responsible AI Principles · Comparing AI Systems · Comparing Ethical Issues

Key concepts:

- The EU AI Act regulates by risk tier — unacceptable, high, limited and minimal — with obligations that scale to the harm the system can cause.
- ISO/IEC 42005:2025 gives an internal, non-certifiable structure for assessing an AI system's impact on individuals, groups and society across its lifecycle.
- Differential privacy is a mathematical guarantee: the output does not reveal whether any single individual's record was in the dataset, tuned by the epsilon budget.
- Comparing AI systems responsibly means scoring them on intellectual property provenance, data privacy posture and environmental cost — not accuracy alone.
- 'Green AI' routes appropriate tasks to small language models, cutting energy and cost against the 'Red AI' default of always using the largest frontier model.
- Explainability trades off against raw accuracy; XAI frameworks such as SHAP and LIME recover interpretability where the decision affects people's lives.

Activity 10 — AI System Comparison Matrix · Slide 65

Objective: Compare AI systems on intellectual property, data privacy and environmental impact (A5).

Scenario: Score ChatGPT, Claude and Gemini across five responsibility dimensions and plot them on a radar chart — turning 'which AI should we use?' into an evidence-based comparison rather than a brand preference.

Duration: 15 minutes.

Activity 11 — Differential Privacy Explorer · Slide 66

Objective: Explain how differential privacy protects individuals and the privacy-utility trade-off it imposes (K1, K3).

Scenario: Turn the epsilon dial and watch the privacy-utility trade-off happen in front of you — strong privacy makes the answer noisy, weak privacy makes it accurate but leaks.

Duration: 15 minutes.

Activity 12 — Explainable AI (XAI) Explorer · Slide 67

Objective: Compare explainability techniques and the accuracy-interpretability trade-off (K3, A6).

Scenario: Open the black box on a loan-approval model: adjust an applicant's profile, watch the decision flip, and compare how LIME and SHAP each explain why.

Duration: 15 minutes.

Activity 13 — The Ethical Paradigms of AI Resource Allocation · Slide 68

Objective: Compare ethical issues in AI applications across competing ethical frameworks (A6).

Scenario: Allocate a scarce resource — 50 treatments among 120 eligible patients — three times over, once under each ethical paradigm, and confront the fact that all three are defensible and they disagree about who lives.

Duration: 15 minutes.

Activity 14 — AI Ethics Case Study Analyzer · Slide 69

Objective: Identify ethical issues in real AI deployments and propose remediation (A6, A1).

Scenario: Analyse documented AI ethics failures — IBM Watson for Oncology, the Apple Card credit investigation, Amazon's recruiting tool, Clearview AI — and propose the remediation that should have been in place.

Duration: 15 minutes.

Resources Required

- Projector/screen and PA system for the trainer's slide deck.
- Whiteboard/flip chart and markers for group activity work.
- Printed or digital copies of the Learner Guide for every learner.
- Internet access and a laptop per learner — all 14 in-class activities are browser-based web apps.
- A free Google Gemini API key (aistudio.google.com/api-keys) for Activities 6, 7, 13 and 14.
- Mobile phones (learners' own) for the SSG digital-attendance QR scans.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 30 minutes, open book.
- Case Study (CS) — one continuous scenario with open-ended tasks, 30 minutes, open book.
- Grading: Competent / Not Yet Competent.
- A minimum of 75% attendance (SSG Digital Attendance record) is required for funding eligibility.
- Learners submit their assessment answers and complete the TRAQOM survey on the LMS at https://lms-tms.tertiaryinfotech.com/.